High Level Architecture

DrivableNet-RT: Real-Time Free Space Segmentation Architecture

Dataset: nuScenes

Input

Input Image (128×256 RGB)

Preprocessing

Resize + Normalize

Encoder

Training Phase

Encoder (CNN)

Conv → ReLU → Pool

Feature Extraction

Skip Connections(Feature Fusion)

Bottleneck

Latent Feature Representation

Pixel-wise Prediction

Decoder

Decoder Upsampling + Skip Connections

Output

Inference Phase

Drivable Mask (Binary)

Classes:
0 = Non-Drivable
1 = Drivable

Loss Function: BCE + Dice Loss

- Lightweight U-Net (~488K parameters)
- Real-time inference on CPU (~15–25 FPS)
- Trained from scratch (no pretrained models)
- Training accelerated using GPU (Kaggle)

Handles:

- Road boundaries
- Shadows
- Occlusions